Chapter 11 — Special Theory of Relativity

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — Frames of reference and Einstein’s postulates

Q.1 Distinguish inertial and non-inertial frames and state the postulates of special relativity.

A frame of reference is a coordinate system used to measure position and time. An inertial frame is one in which Newton’s first law holds: a body with no net force moves with constant velocity. A non-inertial frame is accelerating, so objects appear to accelerate without an applied force. Einstein’s special theory applies to inertial frames.

The two postulates are: (1) the laws of physics are the same in all inertial frames; (2) the speed of light in free space is the same for all observers, regardless of the motion of the source or observer. These postulates force space and time to become relative rather than absolute.

Important Conceptual Short Questions — Topic 1

1. Why is an accelerating car a non-inertial frame?

In the car, objects appear to accelerate backward without a real force acting on them. Newton’s first law does not hold in this accelerating frame, so it is non-inertial.

2. How does the constancy of the speed of light make time relative?

If all observers measure the same speed of light even when moving relative to one another, they cannot agree on measurements of time and distance. Time intervals and lengths must therefore depend on the observer’s motion.

3. Why do the laws of physics being the same in all inertial frames matter?

It means no inertial frame is preferred or absolute. Experiments give the same results in all uniformly moving laboratories, so motion can only be measured relative to another frame.

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Topic 2 — Simultaneity, time dilation and length contraction

Q.2 Explain the relativity of simultaneity, time dilation and length contraction.

Two events that are simultaneous in one frame may not be simultaneous in another moving frame, because the speed of light is finite and the same for all observers. Time dilation means that a moving clock runs slow compared with a clock at rest relative to the events: $t = t_0/\sqrt{(1 - v^2/c^2)}$, where t_0 is the proper time measured by the clock at rest relative to the events.

Length contraction means that the length of an object moving relative to an observer is shorter along the direction of motion: $\ell = \ell_0\sqrt{(1 - v^2/c^2)}$, where ℓ_0 is

the proper length. There is no contraction perpendicular to the motion.

Important Conceptual Short Questions — Topic 2

1. Why is proper time always the shorter time interval?

Proper time is measured by a clock at rest relative to both events. Any moving observer measures a dilated time $t = t_0/\sqrt{(1 - v^2/c^2)}$, which is larger because the denominator is less than 1.

2. Why does length contraction occur only along the direction of motion?

Relative motion affects measurements along the line of motion. Dimensions perpendicular to the motion are unchanged, so a moving object is shortened only in the direction of travel.

3. Why are relativistic effects not noticed in daily life?

Everyday speeds are tiny compared with $c = 3 \times 10^8$ m s^{-1}, so v^2/c^2 is almost zero and the relativistic factor is practically 1.

Topic 3 — Mass variation and mass–energy equivalence

Q.3 Explain relativistic mass variation and Einstein's mass–energy relation.

As the speed of a body increases, its relativistic mass increases: $m = m_0/\sqrt{(1 - v^2/c^2)}$, where m_0 is the rest mass. As v approaches c, the mass grows without limit, so an infinite force would be needed to reach the speed of light. This is why a material body cannot reach c.

Einstein's mass–energy relation is $E = mc^2$. The rest energy of a body is $E_0 = m_0c^2$, and its relativistic kinetic energy is $(m - m_0)c^2$. Mass and energy are equivalent; a small loss of mass can release a very large amount of energy because c^2 is enormous.

Important Conceptual Short Questions — Topic 3

1. Why can a material body not reach the speed of light?

Its relativistic mass increases without limit as v approaches c. Accelerating it further would require infinite force and infinite energy, which is impossible.

2. How does E = mc2 explain the large energy released in nuclear reactions?

A small decrease in mass is multiplied by c^2, which is about 9×10^{16} m^2 s^{-2}. Even a tiny mass loss therefore releases a very large amount of energy.

3. Why is the rest energy of a body not zero even when it is at rest?

Mass itself is a form of energy. A body at rest still has energy $E_0 = m_0c^2$ because of its rest mass.

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Topic 4 — Space-time in relativity

Q.4 Explain the concept of space-time and its role in relativity.

Space-time is a four-dimensional mathematical model that combines the three dimensions of space with time as the fourth dimension. In special relativity, different observers may disagree about distances and time intervals, but they agree on the space-time interval between events. This shows that space and time are not separate absolutes but parts of one structure.

In general relativity, mass and energy curve space-time, and bodies follow the straightest possible paths, called geodesics, in this curved geometry. The bending of starlight during the 1919 solar eclipse confirmed this prediction. Gravitational waves are ripples in this space-time geometry.

Important Conceptual Short Questions — Topic 4

1. Why is space-time not absolute in relativity?

Different moving observers measure different lengths and time intervals. Only the combined space-time interval is invariant, so space and time are relative parts of a single four-dimensional structure.

2. How does curved space-time explain gravity?

Mass and energy curve space-time, and objects move along geodesics in this curved geometry. What we feel as gravity is the motion of bodies following these curved paths.

3. Why was the 1919 eclipse important for relativity?

It allowed astronomers to observe starlight passing close to the Sun. The measured bending of the light agreed with the prediction of curved space-time, supporting general relativity.

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